Forward Propagation in Deep Neural Networks

CS4152 — Deep Learning and Neural Networks

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Outline

Forward Propagation in Deep Networks

- Generalizes single-layer forward propagation to L layers.
- Computes activations sequentially from input to output.
- Uses consistent notation: $Z^{[l]}$, $A^{[l]}$, $W^{[l]}$, $b^{[l]}$.
- Works for both single examples and full datasets.

Forward Propagation for One Training Example

- For layer 1: $Z^{[1]} = W^{[1]}x + b^{[1]}$, $A^{[1]} = g^{[1]}(Z^{[1]})$.
- For layer 2: $Z^{[2]} = W^{[2]}A^{[1]} + b^{[2]}$, $A^{[2]} = g^{[2]}(Z^{[2]})$.
- Continue until layer L.
- Final output: $A^{[L]} = \hat{y}$.

Vectorized Forward Propagation (All Examples)

- Replace x with X: stacked training examples as columns.
- Compute:

$$Z^{[l]} = W^{[l]} A^{[l-1]} + b^{[l]}, \quad A^{[l]} = g^{[l]}(Z^{[l]})$$

- $A^{[0]} = X$, the entire dataset.
- Produces $\hat{Y} = A^{[L]}$ for all examples simultaneously.

The Necessary For Loop

- Vectorization removes loops over training examples.
- But: you must loop over layers 1 through L.
- Pseudocode:

```
A[0] = X
for l = 1 to L:
  Z[l] = W[l] A[l-1] + b[l]
  A[l] = g(Z[l])
```

- This is the only unavoidable loop in a neural network implementation.

Dimension Discipline

- $W^{[l]}$: shape $(n^{[l]}, n^{[l-1]})$
- $b^{[l]}$: shape $(n^{[l]}, 1)$
- $Z^{[l]}, A^{[l]}$: shape $(n^{[l]}, m)$
- Careful dimension tracking reduces implementation bugs.
- Essential for building stable deep network code.

Key Takeaways

- Forward propagation is recursive across layers.
- Vectorization accelerates computation for the full training set.
- A single loop over layers is necessary and expected.
- Dimensional analysis is crucial for correct deep network implementation.